Zachary **Wieand**

Software Engineer · Site Reliability Engineer

zmwieand@gmail.com | zachwieand.com | zmwieand | zachary-wieand

Skills

- **Cloud:** AWS · GCP · EKS · GKE · WAF · Cloudflare
- **Languages:** Go · Python · Bash
- **Infrastructure:** Kubernetes · Helm · Istio · Docker · KEDA
- **CI/CD:** Jenkins · GitHub Actions
- **IaC:** Terraform · Ansible
- **Observability:** Datadog · Prometheus · New Relic · Splunk

Work Experience

ACV Auctions

Buffalo, NY

Staff Engineer - Infrastructure/Technical Lead — *April 2025 - Present*

- Designed a multi-region disaster recovery architecture to protect AWS and Kubernetes workloads against ransomware and regional failures, establishing an RTO of less than 24 hours.
- Architected and implemented Cloudflare at the edge for all public-facing traffic, centralizing security controls to improve DDoS resilience for 500 applications serving 120M requests per day.
- Led the implementation of Kubernetes security standards, developing hardened base images and enforcing secure workload configurations across 25 Kubernetes clusters and 1,000+ workloads, reducing critical vulnerabilities by 40%.
- Architecting the infrastructure for preview and sandbox environments enabling 450+ Product and Engineering employees to rapidly prototype through agentic development.

Senior Site Reliability Engineer/Technical Lead — *Jan 2022 - April 2025*

- Introduced spot instances with Karpenter across EKS clusters, reducing compute costs by $20,000 per month.
- Streamlined application deployment by creating standardized Helm charts enabling developers to autonomously deploy and rollback HTTP APIs, Kafka consumers, CronJobs, and database migrations, achieving 80% workload adoption within the first 12 months.
- Implemented SOX compliance checks into the CI/CD deployment process to enforce audit requirements and track release events.
- Created shared Go modules to encapsulate best practices, including structured logging, HTTP setup, and distributed tracing.
- Developed a multi-agent Jenkins pipeline architecture, reducing pull request pipeline latency by 60%.
- Encapsulated golden signal alert conditions in Datadog into standard application Helm charts, tripling ACV's monitoring coverage.

Site Reliability Engineer III — *Oct 2020 - Jan 2022*

- Migrated several AWS components such as EKS clusters, VPC/networking, WAF, and Security Groups into Terraform modules, eliminating 10+ hours a week of manual configuration work.
- Developed Helm chart linting and testing rules performing Kubernetes schema validation and dependency verification, catching recurring misconfiguration patterns before they reached production clusters.
- Developed a system for Horizontal Pod Autoscaling, using Prometheus and KEDA, to enable 100+ workloads to scale on custom metrics like Kafka consumer lag.

Site Reliability Engineer II — *Apr 2019 - Oct 2020*

- Built and deployed a fluentd application to all Kubernetes clusters to parse and ship logs to Splunk and New Relic.
- Stood up a staging environment for deployment and performance testing, resulting in a 10x reduction in deployment failure rate.
- Developed template repositories following software and CI/CD best practices, enabling teams to deploy to production in less than 15 minutes.

Lockheed Martin

Manassas, VA

Software Engineer Associate, International Submarine Programs — *Aug 2017 - Mar 2019*

- Built Jenkins pipelines to provision complete development and runtime environments across Fedora and RHEL.
- Led migration of software and development environments from legacy Fedora/RHEL releases to RHEL7.
- Containerized application environments with Docker and automated deployment with Ansible, reducing deployment time by 30%.

Google

Mountain View, CA

Software Engineer, Tools and Infrastructure Intern — *May 2016 - Aug 2016*

- Extended an iterative testing tool to support Java regression tests, enabling engineers to execute tests 5x faster.
- Added test case filtering and resolved race conditions and other defects to improve tool performance and release stability.

Education

University at Buffalo

Buffalo, NY

B.S. in Computer Science · Minor in Mathematics — *Aug 2013 - May 2017*